

Scott M. Robertson

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SUMMARY

Analytics-focused Software Engineering Team Lead with experience modernizing legacy Node.js services, migrating large API surfaces from JavaScript callbacks and MySQL-backed logic to TypeScript, async/await, and Snowflake-backed data access patterns. Strong background in endpoint migration, API contracts, DTO interfaces, unit testing, developer enablement, and maintainable backend architecture.

EXPERIENCE

*ClassLink, **Software Engineering Team Lead***

Feb 2025 - Present

- Lead 4 engineers across six backend services and the Angular frontend of ClassLink Analytics; own architecture direction, production releases, and platform reliability.
- Directing the rewrite of the product's core REST API from legacy JavaScript/MySQL to TypeScript and Snowflake, authoring the migration playbook the team applies across ~35 feature domains
- Cut over live event ingestion from direct MySQL writes to an AWS API Gateway pipeline with zero data loss, completing a dual-write migration I designed as an engineer
- Unlocked a new market tier, state education agencies, by designing multi-district query middleware that reused every existing single-district endpoint instead of duplicating the API
- Raised the team's engineering bar by introducing Vitest with 3,600+ lines of controller tests and writing 1,600+ lines of architecture docs that onboard engineers onto legacy services
- Own production reliability for the district-facing platform: zero-downtime audit-store migration, school-hours load-protection windows, hot-path insert batching — eliminated recurring report timeouts

*ClassLink, **Software Engineer***

Nov 2021 - Feb 2025

- Backend engineer on a multi-tenant analytics platform ingesting K-12 usage events for school districts across 18 sharded MySQL hosts, Redis, SQS, and S3.
- - Owned the summarization pipeline powering all ClassLink Analytics dashboards for 3+ years — 55% of every commit in the service's history, nearly 3× the next contributor
- Re-architected the pipeline from a single cron process into an SQS scheduler/worker fleet, horizontally scaling per-tenant batch jobs across 18 database shards and thousands of tenant schemas
- Cut historical backfills from weeks to under one day by replacing raw-log table scans with incremental, resumable aggregation over intermediate tables
- Seeded the company's next-generation data platform by building a streaming MySQL-to-S3 export of TBs of historical logs — resumable+ full production runs
- Root-caused systemic over-counting in customer-facing usage metrics across the sharded fleet, and prevented a production outage by alerting on database ID exhaustion before overflow
- Ran releases for four production promotion — 240+ integration merges across staging, beta, and prod pipelines

*Osmose Utilities Services, **Software Engineer, Software Engineering Intern***

May 2019 - Nov 2021

- Full-stack developer on Osmose 360™, the customer dashboard delivering analytics, reports, and interactive mapping to utility structure owners.
- Led development of an internal application for managing customer projects and accounts, carrying it from concept through delivery
- Built new Vue components across five customer-facing web apps, with unit tests covering both new and existing code
- Extended the ESRI ArcGIS mapping platform to visualize new categories of customer infrastructure data
- Developed service-application management tools end to end in ASP.NET MVC, backed by Azure SQL and Cosmos DB
- Created the interface and backend for uploading engineering design documents and linking them to projects and work requests
- Combined the Google Maps API with US Census data into an interactive data experience, and built grid visualizations with Telerik Kendo UI

EDUCATION

Rochester Institute of Technology, Rochester, NY

*Bachelor of Science • **Game Design and Development, Minor in Philosophy***

GPA: 3.2 - Dean's List Recipient (Fall 2017 - Spring 2018)